

Proposal to Develop a Clinical Practice Guideline on Emotion Regulation:**Final Report**

American Psychological Association

Working Group on Transdiagnostic/Change Process CPG: Emotion Regulation

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Abstract

The Advisory Steering Committee for Development of Clinical Practice Guidelines created a working group to consider the feasibility of, and appropriate methods for, developing an APA clinical practice guideline for interventions that address a transdiagnostic change process, rather than a categorical disorder. Specifically, the working group was asked to determine whether development of a clinical practice guideline focused on emotion regulation was worth pursuing at this time, given the state of the scientific literature, feasibility, and considerations of clinical need and utility.

The working group identified a set of goals for a clinical practice guideline based on the idea that if a guideline could mostly achieve these goals, it would be an advance for the field and worth the investment to try to develop a clinical practice guideline based on a transdiagnostic change process or principle.

Goals:

- 1) Inform providers, patients and their families, payers and other stakeholders about what the empirical data indicate regarding the efficacy of treatments targeting emotion regulation
- 2) Enhance clinical utility of a clinical practice guideline so we learn more than typical guidelines tell us about:

- What works, for whom, and under what circumstances
- What is known about widely-used treatments for which there is not rigorous evidence concerning efficacy
- What is known about change processes or principles underlying the effects of efficacious treatments

After careful consideration of the opportunities and challenges likely to arise in developing a clinical practice guideline on emotion regulation, the working group recommends developing and publishing three review protocols that would lead to a clinical practice guideline with three components:

Review 1. Systematic review of the efficacy data leading to recommendations, following the current best practices to reduce bias (and acknowledgement of biases not addressed with this approach). This would include the Population, Intervention, Comparator, Outcomes, Timing, and Setting (PICOTS) questions to assess applicability based on current best practices, and would include the usual review of literature on harms, patient preferences, etc.

Note: Only in unusual cases when there is a good reason *a priori* to think that a given subgroup will have a unique response to a treatment will subgroup analyses, cohort or other studies (that don't meet best practices to reduce bias) be included. In these cases, careful attention to likely confounds and differences in the analytic approach will need to be considered if these studies are used to guide recommendations.

Review 2. Summary of current knowledge (with appropriate caveats about the quality of evidence) for select treatments that were not included in Review 1, when the treatments are both widely used and studies meet stringent, pre-specified inclusion and exclusion criteria.

Review 3. Summary of current knowledge (with appropriate caveats about the quality of evidence) for literature on change processes or principles based on systematic inclusion and exclusion criteria to determine (observational) studies that rigorously try to account for confounding.

This report outlines the working group's process, rationale for a guideline on emotion regulation, review of challenges likely to arise when developing a guideline of this nature and how recognition of these challenges guided the current proposal. Next the proposal and rationale for the three reviews are discussed, along with consideration of how to manage scope and why the working group believes developing this guideline would be very challenging, but ultimately feasible. We close with recommendations for future research in the hopes that the available literature will one day better match the urgent clinical needs.

Keywords: emotion regulation, change process or principle, clinical practice guideline, transdiagnostic, systematic review

Note. Minor edits to the original report were made prior to sharing this version to PsyArXiv, mainly to update citations, and gendered pronoun use.

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Overview of Working Group Process

Group Charge

The American Psychological Association (APA) has been developing clinical practice guidelines for several years now, and thus far the guideline topics have been focused on specific disorders or conditions based on American Psychiatric Association's (2013) *Diagnostic and Statistical Manual of Mental Disorders* (5th ed.; *DSM-5*) and World Health Organization's (2019) *International Statistical Classification of Diseases and Related Health Problems* (11th ed.; *ICD-11*) categories. The Advisory Steering Committee for Development of Clinical Practice Guidelines, which oversees the initiative, proposed creating a working group to consider the feasibility of, and appropriate methods for, developing an APA clinical practice guideline for interventions that address a transdiagnostic change process, rather than a categorical disorder based on a diagnostic taxonomy. Specifically, the working group was asked to determine whether development of a clinical practice guideline focused on *emotion regulation* was worth pursuing at this time, given the state of the scientific literature and considerations of clinical need and utility. The outcome of this exploration would help determine whether it would be possible to develop a clinical practice guideline on the topic in a manner that meets international best practices (Institute of Medicine, 2011). Ultimately, the working group was tasked with preparing a summary report for the Advisory Steering Committee, including specific recommendations on whether and, if appropriate, how to proceed in developing a clinical practice guideline on emotion regulation, as well as general recommendations on developing guidelines for transdiagnostic processes. The group was asked to determine whether it recommends that APA produce a clinical practice guideline on emotion regulation based on whether that guideline would be clinically useful, address a gap in existing guidelines, and be feasible (in terms of not being too broad and having sufficient, rigorous empirical literature to support systematic review).

Group Formation

A subgroup of Advisory Steering Committee members met to discuss the formation of a working group to consider a transdiagnostic change process clinical practice guideline. The subgroup discussed both the benefits and drawbacks of such a guideline, as well as general clinical utility of clinical practice guidelines. The subgroup concluded that forming such a working group had benefits and determined that at least two current members of the Advisory Steering Committee would serve along with at least five scientific consultants with expertise ranging from emotion regulation theories, treatment, and dissemination to clinical practice guideline development. The subgroup considered numerous relevant topical domains and corresponding experts, and then reached out to experts with a proposal to participate in the working group. Candidates were invited from three main communities: basic science, treatment development, and clinical practice. Although the subgroup did not use a quota approach, attention was paid to representation of expertise across diagnostic categories given the goal of being transdiagnostic, representation of expertise across many treatment orientations given the goal of being transtheoretical, and (some) representation among committee members across demographic characteristics such as gender, age/career stage, race and ethnicity, sexual orientation and gender expression, etc. The invited group members were selected after discussion among the subgroup with additional consultation requested from individuals with expertise in areas not well represented in the group. As potential members accepted and declined the invitation, the subgroup monitored overall diversity and breadth of expertise as they considered further experts to invite to participate to join the project. Rather quickly, eight individuals agreed to participate with two co-chairs. A ninth individual was added to fill a gap once an obvious deficit in content expertise was identified. Each individual completed the conflict of interest disclosure form specific to the clinical practice guideline initiative and began participating in regular videoconference calls.

Meeting Structure

The working group met in person in Washington, DC early in its work in September 2019. At this meeting, the working group received an overview of clinical practice guideline methodologies and held a discussion of concerns raised by the community in response to earlier APA clinical practice guidelines. The group discussed topics pertaining to the scope of the emotion regulation target, the scope of interventions, which population demographic characteristics should be targeted, which diagnostic considerations should be targeted, and the structure of the group's report. In addition to the in-person meeting, the working group has held monthly working calls, with numerous small working groups meeting by phone or videoconference in between the large group calls to further aspects of its work.

Conflict of Interest Review

Before final appointment to the working group, nominees provided information about possible conflicts of interest, a significant issue in the Institute of Medicine (IOM) standards and current best practices in guideline development. *Conflicts of Interest* (COI) are defined as “a divergence between an individual's private interests and professional obligations” (Institute of Medicine, 2011, p. 78; the definition is drawn from Schünemann et al., 2009, p. 565). The IOM report additionally discusses intellectual COI relevant to clinical practice guidelines, defined as “academic activities that create the potential for an attachment to a specific point of view that could unduly affect an individual's judgment about a specific recommendation” (Institute of Medicine, 2011, p. 78; the definition is drawn from Guyatt et al., 2010, p. 739).

Candidates to the working group each completed an APA COI disclosure form: see <https://www.apa.org/about/offices/directorates/guidelines/declaration-development-panel.pdf>. Emphasis was placed on disclosing all potential conflicts for the APA staff and Advisory Steering Committee members to review and decide upon. Although intellectual affiliations were expected, no working group members were to have significant known financial or other conflicts that would compromise their ability (or appearance thereof) to weigh evidence fairly. The

Advisory Steering Committee understood however that some “adversarial collaboration” (Mellers et al., 2001) or standing for different points of view was expected and encouraged as part of the process. Upon successful completion of the reviews, the Advisory Steering Committee and APA staff invited members to join the committee. At the in-person meeting, members verbalized any actual or potential conflicts, so all members of the guideline development panel would be familiar with the diversity of perspectives and range of possible influences and biases. Working group members and staff were asked to give prompt updates if there were any changes in their disclosures that could be relevant to the development of an unbiased working group product.

Rationale for a Clinical Practice Guideline on Emotion Regulation

The rationale for our focus on the *transdiagnostic change process* of emotion regulation is founded on the growing appreciation of transdiagnostic approaches over diagnostic taxonomies, such as *DSM-5* and *ICD-11* (see Dalgleish et al., 2020, for review of current status). Such approaches attempt to identify processes or mechanisms that are critical across multiple clinical conditions. They are consistent with the empirical evidence that supports dimensionality in psychopathology and the course of mental health, and that indicates vast comorbidity among, and heterogeneity within, diagnoses. They have also been advanced to address historical concerns regarding cultural biases in the above-mentioned taxonomies (e.g., Jane et al, 2007; Riquino et al., 2021; Rodriguez-Seijas et al., 2015). Focusing on transdiagnostic change processes provides significant cost-effective advantages with regard to training and dissemination—and subsequently, implementation in healthcare systems. Focusing on critical change processes or principles is often very clinically useful because targeting these change processes may impact multiple problem areas simultaneously (Castonguay et al., 2019). There are several noteworthy transdiagnostic clinical approaches that vary in design and breadth (Barlow et al., 2017; Black et al., 2018; Craske, 2012; Ehrenreich-May & Chu, 2014). Although there are a number of systematic reviews and meta-analyses concerning their effects

(Andersen et al., 2016; Fusar-Poli et al., 2019; Newby et al., 2015; Newby et al., 2016; Pearl & Norton, 2017; Păsărelu et al., 2017; Reinholt & Krogh, 2014), there remain challenges to the development of a transdiagnostic clinical practice guideline –largely due to the predominance of research methodology and clinical trials based on diagnostic taxonomies. We will consider many of these challenges in this report.

The decision to focus on *emotion regulation* in particular is driven in no small measure by a significant shift in emphasis over the past 40 years on emotion in psychotherapy –from something secondary or epiphenomenal to something central or essential in the change process. For example, in the cognitive-behavioral tradition, emotion was once construed as a post-cognitive phenomenon (Beck, 1976) before significant reformulations (Greenberg & Safran, 1987) brought emotion front and center, and highlighted its integral relationship to core processes. We have seen this development in psychoanalytic and humanistic psychotherapies as well (e.g., Jurist, 2019; Greenberg & Paivio, 1997). With this change has come a proliferation of treatment models (with empirical support) in which emotional experiencing and regulation are primary targets of intervention (e.g., Bateman & Fonagy, 2006; Greenberg, 2011; Hofmann, 2016; Kring & Sloan, 2010; Leahy, 2009; Linehan, 2014; Mennin & Fresco, 2013; McCullough, 2003; Muran & Eubanks, 2020; Wonderlich et al., 2015; Young et al., 2003). This progression in clinical practice coincides with and has been fueled by advances in emotion science (see Barrett et al., 2016), especially with regard to emotion regulation, its neurological bases and relation to psychopathology and resilience (see Gross, 2014).

In sum, this initiative's focus on a transdiagnostic clinical practice guideline on emotion regulation is intended to integrate the empirical science on emotion disorders and their treatment with the demands of the field with regard to training, dissemination and implementation.

Emotion Regulation Grid Construction

As we sought to clarify the targets and scope of a clinical practice guideline on emotion regulation, the working group created a grid to organize and document information. Our main objective was to organize the different components of emotion regulation in such a way that it would capture the language and structure commonly used to understand emotion regulation across different communities: basic science, treatment development, and clinical practice. Thus, efforts were made to use language that was inclusive of many different theoretical approaches. The emotion regulation grid is presented in Table 1.

In an ideal world, a clinical practice guideline focused on a transdiagnostic process like emotion regulation would be able to make recommendations at multiple levels. That is, the guideline would draw from the best available research to allow for causal claims tied to the impact of specific therapy techniques, the broader elements of therapy those techniques are intended to implement and be clearly linked to well-established change processes or principles. Moreover, we could do this separately for different domains of emotion regulation (i.e., identifying emotions, monitoring emotions, selecting strategies, implementing strategies; see Gross, 2015), which would allow us to address nuanced questions to closely refine our treatment plan. For example, when considering the case of a client with difficulties in the domain of identifying emotions, we would ask: 1) what is known about the efficacy of attending to and labeling emotions (therapy techniques that fall within the broader category of emotion awareness, a core element of many therapies) and 2) whether those techniques do, in fact, facilitate awareness, with a focused measure showing movement on that particular therapy principle. Unfortunately, the literature is nowhere near that comprehensive, and we have minimal data to allow for systematic reviews at the level of specific therapy principles, elements, or techniques. Instead, we typically have outcome data at the level of treatment packages; for example, did dialectical behavior therapy (DBT: a multi-faceted, long-term therapy that involves dozens of techniques that draw from many therapy elements and change principles) show

efficacy in altering emotion regulation (typically measured with one questionnaire that collapses across different domains of emotion regulation)?

We hope that, in years to come, the literature will allow for more refined conclusions. However, based on our evaluation of the existing literature, we conclude that currently it is necessary to review studies at the level of treatment packages to have sufficient rigorous efficacy literature to draw conclusions. That said, to help move the field toward the desired, more sophisticated understanding of emotion regulation across levels, the working group offers Table 1 as a starting point to help researchers and providers across orientations recognize the links between problems¹ in different emotion regulation domains (drawing from the basic science literature) and the therapy principles, elements, techniques² and packages designed to address those problems. We view this table as a work in progress, with full understanding that other researchers and providers could quite reasonably make different decisions about how the components should be organized. To this end, we hope a future clinical practice guideline development panel will revisit this effort and also invite community input. Our intent was to use inclusive language that allows for related processes and practices across therapeutic orientations to be represented in the table (e.g., we pair the therapy elements of mentalization and cognitive reappraisal, which draw from distinct theoretical and treatment traditions, but share a common focus on the principle of promoting meaning making). In this way, we hope that tables like this can start to bridge basic research and clinical practice by making links more explicit and start to connect different therapy orientations to better recognize shared principles underlying their approaches (while still acknowledging important differences between approaches).

Table 1

Emotion Regulation (ER) Domains (Definitions and Problems) by Therapy Principles, Elements, Techniques and Packages

Relevant ER Domain(s)	Therapy principles	Therapy elements	Therapy techniques	Therapy package
Identification and monitoring				
Definition(s):				
1. Awareness and differentiation of emotion	Facilitate awareness	Exposure to, focus on emotion (<i>bodily felt experience, including body posture, gestures, facial expressions, action urges</i>)	Therapist helps identify and label emotion, focuses on immediate experience, uses empathic reflections and validations of emotional experience (<i>all also apply below</i>)	ACT CBT EFT ERT EDT MBT SE SFT TFP
2. Evaluation whether emotion regulation approach needs to be altered				
Problem(s):				
1. Recognition of subjective and physiological experience of emotions (e.g., what emotion am I experiencing? what is my arousal level?)				
2. Tracking of emotional state and regulation efforts in line with goals in order to guide selection (i.e., to guide adjusting or shifting regulation strategies as needed)				

Relevant ER Domain(s)	Therapy principles	Therapy elements	Therapy techniques	Therapy package
Selection and implementation				
Definition(s):				
1. Selection of emotion regulation strategies to meet the situational demands and	Encourage acceptance	Mindfulness (decentering or distancing)	Therapist facilitates nonjudgmental observation and compassion skills	ACT CBT DBT ERT EFT
2. implementing the strategies	Promote meaning making	Mentalization (<i>reflective function</i>), experiencing level (<i>higher order</i>), cognitive reappraisal, connect emotion to interpersonal context and individual values	Therapist explores emotional experience, offers interpretation (<i>internal/external dynamics: conflicts/ambivalence, defenses/avoidance, beliefs/attitudes, needs/wishes</i>), uses guided discovery (<i>Socratic method</i>), applies two-chair work, facilitates cognitive restructuring (<i>including alternative perspectives</i>), examines how emotions and relationships covary	ACT CAT CBT EFT IPT MBT MBSR SE SFT TFP
Problem(s):				
1. Recognition of strategies needed to adaptively meet demands, with a sufficiently broad repertoire of accessible strategies, and				
2. flexible application of strategies	Increase tolerance	Mindfulness, reappraisal, distraction, modulation	Therapist offers interpretation (see above), regulates with own behavior (coregulation), teaches distress tolerance skills	ACT CBT DBT MBT SFT TFP
	Facilitate behavior change	Situation modification, alternative behavioral response	Therapist facilitates changes in situation and behavior, considers situational demands and available resources, values and goals (<i>homework, experiments</i>), schedules activities that are reinforcing and values-consistent	BA CBT

Note. ACT = Acceptance and Commitment Therapy; BA = Behavioral Activation; CAT = Cognitive Analytic Therapy; CBT = Cognitive-Behavioral Therapy; DBT = Dialectical Behavioral Therapy; EFT = Emotion-Focused Therapy; EDT = Experiential Dynamic Therapy; ERT = Emotion Regulation Therapy; IPT = Interpersonal Psychotherapy; MBSR = Mindfulness-Based Stress Reduction; MBT = Mentalization-Based Therapy; SE = Supportive-Expressive; SFT = Schema-Focused Therapy; TFP = Transference Focused Psychotherapy.

Challenges to Developing a Clinical Practice Guideline on Emotion Regulation

Here we summarize and discuss some of the critical challenges to developing a clinical practice guideline on transdiagnostic change processes in general and on emotion regulation in particular.

Challenge 1: Defining Emotion Regulation: From Basic to Clinical Science

Summary: One of the first challenges in establishing a clinical practice guideline on emotion regulation is variability in how emotion regulation is conceptualized across clinical and non-clinical domains within psychology (e.g., affective science, neuroscience, social and developmental psychology) and even within different clinical traditions (e.g., psychoanalytic/dynamic, cognitive-behavioral, emotion-focused). These conceptual distinctions are critical because they have implications for how clinically significant change and improvement should be defined and measured.

Discussion: In the realm of affective science, emotion regulation is often defined as the activation of a goal to influence the trajectory of an emotional response (Gross et al., 2011). This may involve trying to either increase or decrease experiential, behavioral, and/or physiological aspects of either negative or positive emotion. Sometimes emotion regulation is conscious and effortful, such as when one actively restrains an impulse to yell at someone who has just cut one off while driving. At other times, emotion regulation is engaged outside of conscious awareness (a feature of automaticity), such as when one shifts one's gaze away from an upsetting image online without being aware of the intent to reduce distress. Often emotion regulation involves a person trying to regulate their own emotion, which is referred to as intrinsic emotion regulation. However, emotion regulation may also involve a person regulating another person's emotions, such as when a parent seeks to calm an upset child (Gross, 2015). Inherent in this definition is the ability to attune one's emotional experience and expressions to the ever-evolving social context (Cole et al., 1994).

One framework for understanding the many ways that people try to regulate emotions is the process model of emotion regulation (Gross, 2015). This framework holds that emotions arise in situations that are attended to and appraised in particular ways, leading to a loosely coordinated pattern of experiential, behavioral, and physiological responses that characterize emotion. Thus, emotion regulation occurs when a person identifies the anticipated or actual emotion as a target of regulation, selects a regulation strategy, implements that strategy, and then monitors whether that strategy has led to the desired changes in emotion. According to the process model of emotion regulation, emotion may be regulated at five points in the emotion generative process: (1) selection of the situation, (2) modification of the situation, (3) deployment of attention, (4) change of cognitions, and (5) modulation of responses. Over two decades of research have demonstrated that which strategy a person uses to regulate emotion matters: Different emotion regulation strategies have markedly different affective, cognitive, and social consequences, both immediately and over the longer term (Gross, 2015).

Emotion regulation perspectives stemming from the clinical realm tend to focus on different features of the process model. There is somewhat more (though certainly not exclusive) focus on the explicit features of the model (e.g., changing cognitions) in cognitive behavior therapy approaches, while other clinical approaches emphasize the automatic processes alluded to above, in what is often referred to as affect regulation. An affect regulation perspective frequently sees regulation as being integrated with affect generation (Cozolino, 2010). From this perspective, deliberate regulation of emotion by the client is viewed as necessary, but it is understood as a second-level process (carried out consciously) that is implemented either after the emotion is generated or to consciously prevent it from being generated (e.g., situation selection or modification). The primary processing is believed to occur at an unconscious level, and this automatic or implicit regulation is seen as being achieved at the point of generation. For example, one might internalize the empathic soothing of the other so that the self feels more secure and emotion is less likely to be activated; or one might

transform emotions by changing how they are processed (at the point of generation) so that the emotion never arises, or it does so in an already regulated state. Consistent with this notion, Berking et al. (2008) demonstrated the importance of implementing emotion regulation skills as a treatment target in psychotherapeutic interventions, but they also speculated that there might be automatic processes involved in managing negative affect and that interventions addressing implicit or unconscious, intra-psychic regulating mechanisms are necessary. Thus, affect regulation is an important aspect of an encompassing definition of a transtheoretical, transdiagnostic perspective of emotion regulation.

Across these and other perspectives is an appreciation of the role that regulation of emotions (either deliberately or implicitly) plays in well-being and ill-being. On the well-being side, the ability to enact emotion regulation strategies in a manner consistent with cultural norms is often considered a part of healthy individual functioning (Cole et al., 1994), as well as a core feature of healthy relationship functioning (Soto et al., 2013). On the ill-being side, failure to comply with expected emotional norms can be problematic, just as implementing certain strategies to regulate emotions (e.g., violent outbursts) can also pose problems for the individual and those around them. When a pattern of emotion regulation (or lack thereof) becomes linked with impaired functioning for an individual, this is often referred to as emotion dysregulation (Cole et al., 1994; Gratz & Roemer, 2004; Gross & Jazaieri, 2014). Similarly, an individual's attempt to avoid, minimize or convert intolerable negative affect can also be seen as leading to emotion dysregulation, especially in more severe pathologies. Thus, dysregulation can be understood as a tendency for individuals to avoid regulating or to regulate emotions (unconsciously or consciously) in a way that impairs their short and/or long-term functioning. Not surprisingly, emotion dysregulation is a common feature across many psychological disorders. It follows that clinical science and research are often more concerned with the nature and definition of emotion dysregulation, rather than the definition of emotion regulation per se (Gratz & Roemer, 2004; Tull & Aldao, 2015). Accordingly, many treatments of psychopathology target

emotion dysregulation (either explicitly or implicitly) as a change process or mechanism and, less frequently, measure the extent of dysregulation as a treatment outcome.

Importantly, determining when dysregulation occurs, or whether and when a given emotion regulation strategy is harmful, is defined by the outcome of the regulatory process. In other words, the (mal)adaptiveness of any form of emotion regulation tends to be context dependent (Aldao, 2013; Bonanno & Burton, 2013; Gross, 2015), and thus there is no straightforward way of identifying optimal emotion regulation *a priori*. Differentiating adaptive versus maladaptive patterns of emotion regulation often requires a thorough functional analysis of patients' behaviors to identify how regulation impacts their emotional and physical well-being (Aldao & Christensen, 2015). For example, violent outbursts would, in most circumstances, be considered problematic, but they might confer some advantage to a given individual who is in a threatening situation. Further, it becomes crucial to differentiate short- and long-term consequences, because many forms of regulation might produce divergent outcomes across time, thus making the differentiation between adaptive and maladaptive regulation even more challenging. For example, consuming several drinks of alcohol at a work party to lower anxiety might be effective in the very short term, but might lead to detrimental consequences in the long term (e.g., tarnished reputation with colleagues). Lastly, from the standpoint of interventions, clinicians sometimes encourage clients to use emotion regulation strategies that might be less than optimal, but that might nonetheless represent an important stepping-stone for that patient. For example, a clinician doing dialectical behavior therapy would encourage a patient to engage in distraction versus self-harm, with the goal of eventually being able to sit with tough emotional experiences ("surf the wave") rather than avoid them.

Implication for Current Proposal

Not only are there important distinctions in how emotion regulation is defined and examined across clinicians, but the contextual factors involved in understanding whether emotion regulation is being positively or effectively changed make it difficult to arrive at general

conclusions about what does and does not work when considering interventions that target these processes. This concern was part of the motivation for the working group to create Table 1, so we could begin to build a shared framework and language to recognize different domains of emotion regulation and their associated clinical interventions. Moreover, recognition of this challenge and the need to manage the scope of a first clinical practice guideline on a multifaceted change process like emotion regulation led to the decision that this proposal would focus on emotion regulation, rather than emotion generation. This was a difficult decision, and the working group had some different perspectives on how to manage the scope and definitional challenges; however, the group settled on this focus on emotion regulation for this proposal for an initial clinical practice guideline on emotion regulation, with the hope that future guidelines could focus more directly on emotion generation.

Challenge 2: Existing Literature Does Not Adequately Represent the Treatments Being Delivered or Diverse Populations We Treat

Summary: High-quality efficacy trials tend to focus on a limited set of therapeutic approaches and do not include sufficient well-powered studies with diverse populations, particularly those with marginalized identities that are traditionally underserved.

Discussion: A clinical practice guideline is necessarily limited by the literature available to review. There are at least two key areas where this limitation is concerning for a guideline on interventions for emotion regulation. The first is the widely recognized lack of well-powered studies that include samples with diverse identities, especially in terms of racial and ethnic (as well as diverse gender and other) identities. This means we lack critical data about whether treatments have comparable effects (i.e., show treatment homogeneity versus heterogeneity of effects) for many marginalized and underserved populations. This problem is in no way unique to studies of interventions for emotion regulation but remains a serious concern for the field and likely contributes to the many disparities observed in healthcare quality and access.

A second area where lack of diversity is problematic concerns the insufficient representation of different treatments in the rigorous randomized controlled trial (RCT) literature that is typically the main source of studies for systematic reviews. For example, although depression is among the most extensively studied disorders (Cuijpers et al., 2008), a substantial body of the research base focuses on cognitive-behavioral therapy (APA, 2019). A smaller portion of the research base focuses on other treatments that are broadly used but perhaps less likely to be studied using well-powered RCTs, such as psychodynamic therapy, emotion-focused therapy, and more (APA, 2019). This means clinical practice guidelines often only consider a limited number of treatments, so providers, consumers and other stakeholders know little about the status of many widely used treatment approaches. It is critical for the field to produce more high-quality evidence investigating a broader range of different psychotherapies (APA, 2019) and also important that people learn all we can from the available treatment literature.

Implication for Current Proposal

The working group spent a great deal of time considering how to manage these significant challenges given our desire to be more inclusive in the evidence reviewed and provide more information to providers, consumers, and various stakeholders while simultaneously not making recommendations that had high risk of being biased. This led to our decision to include a very small set of subgroup analyses to look at treatment heterogeneity as a function of different identities as part of Review 1 when those analyses were identified *a priori* and selected studies followed systematic inclusion and exclusion criteria. In addition, we elected to include an additional review (Review 2) that is not typically part of a clinical practice guideline to describe the current state of the evidence for treatments that are widely used but where the efficacy studies did not meet the rigorous criteria expected for Review 1 (below, see Proposal: Three-Review Approach to Develop a Clinical Practice Guideline on Emotion Regulation).

Challenge 3: Literature is Not Organized by Change Processes

Summary: The literature on intervention efficacy has mainly been based on RCTs, which have been typically defined by diagnostic taxonomies, rather than transdiagnostic change processes. Moreover, much of the research on change processes is based on observational designs involving correlational analyses.

Discussion: There have been several considerations over the years regarding change processes (Muran & Lipner, 2023). These have been described in terms of “common factors,” such as the therapeutic alliance and patient expectation, but also include treatment specific tasks (Frank, 1961; Lambert, 1986; Orlinsky & Howard, 1987; Rozenzweig, 1936; see meta-analyses by Wampold, 2015; Wampold & Imel, 2015). They have also been described in terms of “relationship factors” by an APA Division 29 Task Force that conducted a series of meta-analyses on the alliance, group cohesion, goal consensus, empathy, positive regard, genuineness, expression, and rupture repair, to highlight a few factors (Norcross & Lambert, 2019). More broadly, they have been described as “principles of change” (POCs: Goldfried, 1980); accordingly, Castonguay and colleagues (2019) organized the empirical evidence on POCs as follows:

- a. *Client prognostic principles:* characteristics that correlate with improvement
- b. *Client moderating principles:* characteristics that influence intervention efficacy
- c. *Client process principles:* patient behaviors that affect improvement
- d. *Therapy relationship principles:* patient-therapist interactions that affect improvement
- e. *Therapist intervention principles:* therapist behaviors that affect improvement

Another approach to change processes (specifically aimed to address concerns about narrow diagnostic taxonomies that do not account for comorbidity) has been described in terms of “transdiagnostic processes.” Barlow and colleagues (e.g., Barlow & Farchione, 2017) proposed a prominent example with their *Unified Protocol* (UP), based on empirically-supported

cognitive-behavioral and acceptance-based therapies for emotional disorders. This treatment protocol aimed to distill the main elements from various therapy approaches that brought about change for mood, anxiety, and other emotional disorders. Research has demonstrated the efficacy of the UP in *causing* symptomatic reduction and therefore overall change (Barlow et al., 2017; Barlow et al., 2020; Boswell et al., 2014; Sauer-Zavala, Fournier et al., 2021; Sauer-Zavala, Rossellini et al., 2021). In addition to the promise of this transdiagnostic protocol, there have been other efforts to define transdiagnostic processes (Harvey et al., 2004) with growing empirical support (Dalgleish, et al., 2020; Newby et al., 2015), including efforts regarding emotion regulation (Kring & Sloan, 2010) and process-based therapy (e.g., Hofmann & Hayes, 2019). In another noteworthy initiative, the Human Behaviour Change Project (HBCP: <https://www.humanbehaviourchange.org/>) applied Artificial Intelligence to search publication databases for information about links between behavior change techniques and their mechanisms of action and triangulate such information with expert consensus. The initiative resulted in a tool that explores the relationship between 74 behavior change techniques and 26 mechanisms of action.

Research on intervention efficacy has mostly been based on experimental RCTs with inclusion criteria predominantly defined by the diagnostic taxonomies of DSM and ICD. Though the RCT is reasonably recognized as the gold standard to test intervention cause and effect, its predominant application based on a diagnostic taxonomy has resulted in an unfortunate dearth of rigorous, experimental research on transdiagnostic change processes. Instead, much of the research on change processes is based on observational studies (e.g., cohort and case control) involving correlational analyses, though there has been limited but increasing consideration of temporal precedence and causality with regard to treatment outcome (e.g., Barber et al., 2021). This has resulted in methodological innovation over the years, including the application of mediational analyses to define “change mechanisms” (Kazdin, 2007; Laurenceau et al., 2007) and the question of causality has also been addressed by the application of mixed methods

(incorporating qualitative strategies) to define “change events” (e.g., Greenberg & Newman, 1996; Safran & Muran, 1996), as well as reconsidered in exclusively qualitative method (Maxwell & Levitt, 2023).

Implication for Current Proposal

The working group felt it was important to discuss the research on change processes more thoroughly and systematically than typically occurs in a clinical practice guideline, yet also recognized that because many of the relevant studies were not RCTs that would allow for clear causal inferences, these studies would not end up being included in a typical clinical practice guideline efficacy review. This helped guide our decision to include an additional descriptive review (Review 3, focused on change processes; see proposal below) as part of the clinical practice guideline.

Challenge 4: Identifying Transdiagnostic Relevant Populations

Summary: There is no clear consensus amongst clinicians and researchers on how to identify people with emotion regulation problems (i.e., diagnostically versus psychometrically). Emotion dysregulation has been considered with regard to multiple interrelated domains (e.g., self-identity, interpersonal disturbance, impulsive behavior).

Discussion: Identifying populations with emotion regulation problems is complicated. First, there is no clear consensus amongst clinicians and researchers on how to classify people with emotion regulation problems. The *DSM* and *ICD* do not list a disorder of emotion dysregulation. Consequently, only a few clinical studies have selected patient populations based on emotion dysregulation criteria. The absence of precise inclusion/exclusion criteria for selecting relevant populations is problematic. Because emotion and emotion regulation are broad constructs, it is a challenging task to achieve construct specificity. Questions about which populations to include in practice guidelines require choices and a clear rationale about which aspects of emotion and emotion regulation are deemed most important. Individuals with emotion regulation difficulties are ubiquitous and seen across clinical settings. Emotion regulation

difficulties are implicated in most mental health problems and specific disorders. Some clear examples include post-traumatic stress disorder, obsessive compulsive disorder, bipolar disorder, depression, borderline personality disorder, substance use disorder, and disruptive/impulse-control/conduct disorders. Further, there is the complicating issue that emotion regulation problems vary by degree from high functioning to poorer functioning populations. An overwhelming number of populations could be included in practice guidelines on emotion dysregulation. This means including both populations who meet and who do not meet criteria for a disorder. If emotion regulation practice guidelines are to be sensible and practical, the selection of populations that reflect emotion dysregulation needs to avoid being too narrow or overly broad. Further, which populations (and studies) to identify is complicated by the fact that emotion dysregulation is often not well-defined or measured in psychosocial treatment research (as described above). Commonly, research on the treatment of emotion dysregulation and observable behaviors for change are reflected within interrelated domains such as self-identity, interpersonal disturbance and impulsive behavior. This poses challenges in deciding which populations and studies are more or less central to emotion dysregulation.

Implication for Current Proposal

Recognition of this challenge led the working group to decide to include populations at both diagnostic and sub-clinical severity levels and to include a broad range of problem areas, including both *DSM* and *ICD* disorders but also problems in living, such as procrastination and suicidal behaviors, that are intimately tied to difficulties with emotion regulation. Moreover, to recognize the connections across problem areas while also managing scope of a guideline (and avoid treating all problems of emotion regulation as though they are the same), a clustering approach is proposed to meaningfully group different populations (see Managing Scope by Clustering Target Populations below).

Challenge 5: Developing Inclusive, Feasible Search Terms for a Systematic Review

Summary: Conducting a systematic review on interventions that target emotion regulation poses a number of challenges, specifically with regard to search strategies and screening criteria.

Discussion: Systematic reviews may be used to address questions related to the effectiveness of interventions, which can inform the development of clinical practice guidelines or professional practice guidelines. A guideline focused on emotion regulation could address multiple questions, including: 1) What are the effects of psychological interventions on emotion regulation outcomes? 2) What are the effects of interventions designed specifically to improve emotion regulation on emotion regulation outcomes? To answer either question, one would conduct a highly sensitive literature search to identify potentially relevant citations (see Rethlefsen et al., 2022) and then apply prespecified screening criteria to identify relevant studies. Electronic literature searches can be used to identify citations in which specified terms appear in titles and abstracts or in indexed fields (e.g., MeSH terms), but it is usually not possible to search full texts of journal articles and other sources.

Current best practice is to *not* restrict searches and screening using study outcomes. Electronic search strategies typically include several parts or “filters.” A filter might include many terms used to describe the same variable, which are combined using the Boolean operator “or”. Multiple “filters” might be combined using the operator “and” to identify the overlapping citations of greatest relevance. Search strategies commonly include filters for interventions (e.g., psychodynamic psychotherapy), problems and/or populations (e.g., adults with depression), and study types (e.g., RCTs). Because many study outcomes are not mentioned in citation titles and abstracts, limiting searches to citations that mention specific outcomes can miss eligible studies. And because outcomes with small or non-significant results may not be mentioned in titles or abstracts and might not be reported in publications at all (i.e., reporting bias), electronic

literature searches and preliminary screening based on reported outcomes can introduce bias in the review process.

An electronic literature search might be unable to identify all eligible studies that included a measure of emotion regulation. Although studies of many psychological interventions have assessed emotion regulation as secondary or exploratory outcomes, it might not be possible to develop a highly sensitive electronic literature search for such studies. That is, when emotion regulation is not the primary outcome in a study (as might be the case in studies about the effectiveness of psychological interventions for people meeting diagnostic criteria for particular problems), emotion regulation terms might not appear in corresponding titles, abstracts, or indexed terms. Thus, an electronic literature search for all studies that assessed the effects of interventions on emotion regulation might miss many potentially relevant citations, especially those with non-significant results on emotion regulation outcomes. Moreover, a search for studies of any intervention related to emotion regulation would produce a very large number of citations for screening and assessment, which might be practically unmanageable.

Implication For Current Proposal

To manage the potentially enormous number of results that would follow from searching on every potential treatment for a problem area, we elected to limit the searches to *psychosocial therapies* that are specifically designed to target emotion regulation processes. This is because it is more feasible to identify and summarize the results of studies that assess interventions to improve emotion regulation (Question 2 above). Studies of interventions designed to change emotion regulation may be likely to have used terms related to emotion regulation in their titles, abstracts, and other searchable fields. A search for studies of interventions to improve emotion regulation would be narrower than a search for all studies of psychological interventions. Although it might be impossible to eliminate risk of publication bias and selective results reporting, searching for terms related to the intervention, rather than the reported outcomes, would reduce risk of reporting bias in the review. To this end, the working group generated an

initial list of therapies (noted in Table 1) that the group felt would fit the goal of identifying interventions designed specifically to improve emotion regulation on emotion regulation outcomes. A clinical practice guideline development panel would want to review this list carefully and solicit community and expert input before finalizing the list, while keeping in mind the need to have a body of efficacy data that included emotion regulation outcome measures to warrant including the therapy as part of the search terms.

Challenge 6: Identifying Measures of Emotion Regulation Used in Outcome Research

Summary: The challenge here is due to the limitations in the number and nature of outcome measures that target emotion regulation that have been used in intervention studies.

Discussion: As previously discussed, there are multiple definitions of emotion regulation, making it challenging to select what should count as an emotion regulation outcome measure that would be appropriate to include in the reviews that make up a clinical practice guideline. The working group recommends using Gross and colleagues' (2011) approach that emotion regulation involves the *activation of a goal to influence the trajectory of an emotional response*. This approach has the advantage that it is not restricted to one theoretical approach or one specific set of emotion regulation strategies, making it an inclusive approach to what measures could be included, which is advantageous because it allows us to learn from the available literature and not exclude lots of potentially relevant studies. At the same time, it is not so broad that anything counts; the approach implies that some kind of objective marker is needed that there is a goal to change an emotion.

In theory, the measure could be indirect and demonstrate implicit emotion regulation or could be a behavioral indicator, but these measurement approaches for emotion regulation goals are not yet captured well or widely in the literature, so in practice we expect almost all measures used in the studies included in reviews will be questionnaire-based (with a few observer-rated). Notably, symptom measures or simple affect reports (like the Positive and Negative Affect Schedule [PANAS, Watson et al., 1988]) would not be included in our

recommended list of emotion regulation outcome measures because it is not possible to infer that a goal to change the emotion was evident on these measures (versus this being an outcome that happened to occur). Analogously, activities that happen to make us happier, like eating ice cream, would not in and of themselves count as a behavioral indicator of an emotion regulation outcome measure because it is not clear how the presence of an underlying emotion regulation goal would be determined. Admittedly, these distinctions can be challenging to make in practice. This is also consistent with our focus in the current guideline recommendations on emotion regulation versus on emotion generation. As noted, we hope that a future guideline that includes emotion generation will be possible and also hope that we will have a broader range of measures to draw from (including those that reliably capture implicit emotion regulation), but our current recommendations are necessarily limited by the existing measurement and treatment outcome literature.

Implication for Current Proposal

To help identify the measures that could be included as markers of emotional regulation change, we present Table 2 as a starting point for a clinical practice guideline development panel, while fully acknowledging that drawing distinctions among measures to determine whether they show activation of a goal to influence the trajectory of an emotional response is inexact and this list is not comprehensive. The list was prepared by reviewing numerous articles included in prior reviews of emotion regulation measures and treatment studies and noting what outcome measures they used. Each candidate measure was then reviewed by at least three members of the working group to judge whether it met our criterion regarding demonstrating activation of a goal to influence the trajectory of an emotional response. The measures included in Table 2 were selected by consensus among the three judges. (Note the measures, as summarized in Table 2, are primarily self-report and thus privilege explicit processes, some directly and others indirectly related to emotion regulation.)

Table 2*Outcome Measures of Emotion Regulation*

Measure(s)	Scale(s)	Method
Acceptance and Action Questionnaire - II (AAQ-II; Bond et al., 2011)	Experiential avoidance (7 items total)	Self-report
Affective Control Scale (ACS; Williams et al., 1997)	Anger Depressed mood Anxiety Positive (42 items total)	Self-report
Affect Regulation Scale (ARS; Yeo & Frederiks, 2011)	Affective regulation (4 items total)	Self-report
Brief Experiential Avoidance Questionnaire (BEAQ; Gámez et al, 2014)	Experiential avoidance (15 items total)	Self-report
Child Response Style Questionnaire (CRSQ; Meiser-Stedman et al., 2007)	Rumination Distraction Problem-solving (25 items total)	Self-report
Children's Sadness Management Scale (CSMS; Zeman et al, 2001)	Inhibition Regulation coping Dysregulated-expression (12 items total)	Self-report Observer-rated
Children's Worry Management Scale (CWMS; Zeman et al., 2010)	Inhibition Dysregulation Coping (10 items total)	Self-report Observer-rated

Measure(s)	Scale(s)	Method
Cognitive Emotion Regulation Questionnaire (CERQ; Garnefski et al, 2001)	Self-blame Acceptance Focus on thought/rumination Positive refocusing Refocus on planning Positive reappraisal (36 items total)	Self-report
COPE Inventory (COPE; Carver et al., 1989)	Active coping Planning Suppression of competing activities Restraint coping Seeking social support for instrumental reasons Seeking social support for emotional reasons Positive reinterpretation & growth Acceptance Turning to religion Focus on & venting of emotions Denial Behavioral disengagement Mental disengagement (52 items total)	Self-report
Coping Inventory for Stressful Situations (CISS; Cosway et al., 2000)	Task Emotion Avoidance (Distraction, Social Diversion) (48 items total)	Self-report
Distress Tolerance Scale (DTS; Simons & Gaher, 2005)	Tolerance Appraisal Absorption Regulation (15 items total)	Self-report

Measure(s)	Scale(s)	Method
Difficulties in Emotion Regulation (DERS; Gratz & Roemer, 2004)	Nonacceptance of emotional responses Difficulty engaging in goal-directed behavior Impulse control difficulties Lack of emotional awareness Limited access to emotion regulation strategies Lack of emotional clarity (36 items total)	Self-report
Drexel Defusion Scale (DDS; Forman et al., 2012)	Cognitive defusion (10 items total)	Self-report
Emotional Approach Coping Scales (Stanton et al., 2000)	Emotional processing Emotional expression (8 items total)	Self-report
Emotion Regulation Checklist (ERC; Shields & Cicchetti, 1997)	Lability/Negativity Regulation Observations (24 items total)	Observer-rated

Measure(s)	Scale(s)	Method
Emotion Regulation Skills Questionnaire (ERSQ; Grant et al., 2018)	Awareness Sensations Clarity Understanding Modification Acceptance Tolerance Readiness to confront distressing situations when necessary to attain personally relevant goals Self-support (27 items total)	Self-report
Emotion Regulation Questionnaire (ERQ; Gross & John, 2003)	Cognitive reappraisal Expressive suppression (7 items total)	Self-report
Experiencing Scale (EXP; Klein et al., 1969)	Levels of emotional involvement (7-point rating)	Observer-rated
Five-Facet Mindfulness Questionnaire (FFMQ; Baer et al., 2006)	Observe Describe Act with Awareness Nonjudge Nonreact (39 items total)	Self-report
Generalized Expectancies for Negative Mood Regulation Scale (GENMR; Catanzaro & Mearns, 1990)	Behavior and cognitive expectancies for negative mood regulation (30 items total)	Self-report

Measure(s)	Scale(s)	Method
Kentucky Inventory of Mindfulness Skills (KIMS; Baer et al, 2004)	Observing Describing Acting with awareness Accepting (or allowing) without judgment (39 items total)	Self-report
Life Problems Inventory (LPI; Rathus et al., 2015)	Confusion about self Impulsivity Emotion dysregulation Interpersonal chaos (60 items total)	Self-report
Mentalization Questionnaire (MZQ; Hausberg et al., 2012)	Refusing self-reflection Emotional awareness Psychic equivalence mode Regulation of affect (15 items total)	Self-report
Mentalization Scale (MentS; Dimitrijević et al., 2018)	Self-related mentalization Other-related mentalization Motivation to mentalize (28 items total)	Self-report
Mindful Attention Awareness Scale (MAAS; Brown & Ryan, 2003)	Mindful attention awareness (15 items total)	Self-report
Multidimensional Experiential Avoidance Questionnaire (MEAQ; Gámez et al., 2011)	Procrastination Distraction/suppression Behavioral avoidance Distress endurance Repression/denial Distress aversion (62-items total)	Self-report

Measure(s)	Scale(s)	Method
Observer-rated measure of affect regulation (O-MAR; Watson & Prosser, 2004)	Awareness/labeling Modulation of arousal/experience Modulation of expression Acceptance of experience Reflective of feelings/experience (35 items total)	Observer-rated
Philadelphia Mindfulness Scale (PHLMS; Cardaciotto et al., 2008)	Awareness Acceptance (20 items total)	Self-report
Reflective Functioning (RF; Fonagy et al., 1991)	Mentalization (9-point rating)	Observer-rated
Reflective Functioning Questionnaire (RFQ; Fonagy et al., 2016)	Certainty about mental states Uncertainty about mental states (46 items total)	Self-report
Response Style Questionnaire (RSQ; Nolen-Hoeksema et al., 1993)	Ruminative Distracting (32 items total)	Self-report
Responses to Positive Affect (RPA; Feldman et al., 2008)	Emotion-focus Dampening Self-focus (17 items total)	Self-report
Ruminative Response Scale (RRS; Nolen-Hoeksema & Morrow, 1991)	Focus on self Focus on symptoms Focus on mood (22 items total)	Self-report
Savoring Beliefs Inventory (SBI; Bryant, 2003)	Anticipating Savoring the moment Reminiscing	Self-report

Measure(s)	Scale(s)	Method
	(30 items total)	
Social Problem-Solving Inventory – Revised (SPSI-R; D’Zurilla et al., 2002)	Positive problem orientation Negative problem orientation Rational problem solving Problem definition and formulation Generation of alternative solutions Decision making Solution implementation and verification Impulsivity/carelessness style Avoidance style (52 items total)	Self-report
Spielberger State-Trait Anger Expression Inventory-2 (STAXI-2; Spielberger, 1988)	Anger expression Anger control Anger experience (57 items total)	Self-report
Toronto Alexithymia Scale - II (TAS-II; Bagby et al., 1994)	Difficulty identifying feelings Difficulty describing feelings Externally oriented thinking (20 items total)	Self-report
Toronto Mindfulness Scale (TMS; Lau et al., 2006)	Curiosity Decentering (13 items total)	Self-report
Trauma-Related Rumination Items (Meiser-Stedman et al., 2014)	Rumination and perseverative thinking (3 items total)	Self-report
Ways of Coping Checklist (Vitaliano et al., 1985)	Problem-Focused Scale Seeks Social Support Scale Blamed Self Scale Wishful Thinking Scale (68 items total)	Self-report

Proposal: A Three-Review Approach to Develop a Clinical Practice Guideline on Emotion Regulation

In this section, we outline our proposed approach to develop a transdiagnostic clinical practice guideline on emotion regulation. Specifically, we propose that a clinical practice guideline on emotion regulation would include three review components, each of which would be based on a pre-specified and published protocol:

Review 1. Systematic review of the efficacy data leading to recommendations, following the current best practices to reduce bias. This would include the PICOTS questions to assess applicability based on current best practices and would include the usual review of literature on harms, patient preferences, etc.

Note: Only in unusual cases when there is a good reason *a priori* to think that a given subgroup will have a unique response to a treatment will subgroup analyses, cohort or other studies (that don't meet best practices to reduce bias) be included. In these cases, careful attention to likely confounds and differences in the analytic approach will need to be considered if these studies are used to guide recommendations.

Review 2. Summary of current knowledge (with appropriate caveats about the quality of evidence) for select treatments that were not included in Review 1, when the treatments are both widely used and studies meet stringent, pre-specified inclusion and exclusion criteria.

Review 3. Summary of current knowledge (with appropriate caveats about the quality of evidence) for literature on change processes or principles, based on systematic inclusion and exclusion criteria to determine (observational) studies that rigorously try to account for confounding.

Rationale for Recommending a Guideline that Contains Three Reviews

The decision to propose multiple reviews for the guideline follows from the goals driving our working group's efforts to determine a feasible and clinically useful approach to develop a transdiagnostic clinical practice guideline on emotion regulation. Importantly, each review would be systematic in that clear scoping guidelines, inclusion/exclusion criteria for studies, and planned search terms would be established *a priori*.

Goals

1. Inform providers, patients and their families, payers and other stakeholders about what the empirical data indicate regarding the efficacy of treatments targeting emotion regulation
2. Enhance clinical utility of a clinical practice guideline, so we learn more than typical guidelines tell us about:
 - a. “What works, for whom, and under what circumstances” (Paul, 1967; Roth & Fonagy, 2006)
 - b. What is known about widely-used treatments for which there is not rigorous evidence concerning efficacy
 - c. What is known about change processes or principles underlying the effects of efficacious treatments

Efforts to achieve these goals quickly run into barriers tied to the many gaps in the available literature, as outlined in the first half of this report. We thus outline how the proposed three-review proposal may help provide a reasonable framework to balance the desire to be inclusive regarding sources of evidence (so we learn from the available empirical literature) with maintaining rigor of the review process (so we don't make unhelpful recommendations due to including studies with high risk of bias).

Rationale for Review 1's Focus on Rigorous Efficacy Data with Limited Subgroup

Analyses

Review 1 would look most like what is included in APA's (2017, 2018, and 2019) existing clinical practice guidelines in that this systematic review of the efficacy data would follow current best practices to reduce bias (e.g., based on guidelines from the National Academy of Medicine, formerly Institute of Medicine [IOM, 2011]) and would lead to recommendations with different levels of support (e.g., strongly recommend, conditionally recommend, etc.). This review would also include the PICOTS questions to assess applicability based on current best practices and would include the usual review of literature on harms, patient preferences, etc.

Addressing the PICOTS questions to understand moderators of treatment effects has been especially challenging in current clinical practice guidelines, and it has been disappointing that panels are typically not able to draw strong conclusions about moderators given the limited literature. Thus, the working group spent a lot of time considering whether we could do this more effectively in this proposal, consulting with multiple methodologists and with researchers who study treatment outcomes with marginalized and underserved samples. After all, the need to learn “what works, for whom, and under what circumstances” is clear and especially pronounced in light of the systematic disparities in access to high quality care encountered by individuals with various marginalized identities and our field's history of conducting research with samples that do not represent the full diversity of our communities. Yet, our consultation also made salient why we need to be very cautious in drawing conclusions about what works for whom when we do not have a body of good quality, well-powered data to guide the answers.

To make this challenge concrete, imagine Treatment A is recommended as efficacious, but there are insufficient data to determine whether treatment heterogeneity exists as a function of different patient characteristics. This is a very common occurrence; for instance, APA's (2017) clinical practice guideline recommends prolonged exposure for PTSD, but we know little about whether there are different results for Black versus White patients. Naturally, providers

wonder whether to use prolonged exposure with their Black patients given few Black patients were included in some of the key trials used to guide the recommendations. This is a serious problem with our available literature that needs to be addressed.

However, if we try to address this problem with the literature (i.e., the absence of research studies with diverse populations that would enable well-powered, pre-specified tests of subgroup analyses) by including studies with high risk of bias to guide recommendations, it is very plausible that we will end up depriving patients of a beneficial treatment. For instance, if we include a small cohort study of prolonged exposure conducted with Black patients, there is a substantial risk of bias – that study may lack power to find replicable effects, and there is a real likelihood of differences in the way that study was conducted and/or analyzed relative to the larger trials, confounding our ability to compare the results of the cohort study with the larger trials that guided the initial recommendation.

As just one example, perhaps the cohort study was run in a community mental health center while the larger trial was run in multiple psychiatric clinics. The setting thus differs across the studies, but we are focused on the race differences across the samples. If the study restricted to Black patients shows beneficial effects, we are in the same place we were earlier (i.e., we recommend prolonged exposure for PTSD and state there is no reason to make different recommendations based on race of the patient based on the existing empirical literature). Note there may of course be other reasons to make a different treatment choice given a particular patient's values, preferences, prior treatment history, etc., but this is distinct from the recommendations based on a systematic review of the nomothetic literature.

The challenge arises if the study restricted to Black patients does NOT show beneficial effects. Now, if we make a recommendation that there is NOT support for prolonged exposure for Black patients, we can easily end up depriving Black patients of high-quality care when the difference in results across studies was due to random error, differences in comparison groups, differences in sampling or other forms of bias, differences in settings, or one of many other

differences across the studies besides the patients' race. We thus end up exacerbating health disparities when we started with the intention of reducing them.

This leads to the conclusion that we should be very careful not to suggest there are differences across subgroups unless we have really good data to guide that conclusion. However, IF there is a good reason *a priori* to think that a given subgroup will have a different response to a treatment, then it can make sense to include cohort studies and/or subgroup analyses if careful attention is paid to power, and likely confounds and differences in the analytic approach in light of the potential for unintended negative consequences. This would not be the norm though and would need to be done with considerable caution.

Thus, we suggest that the clinical practice guideline development panel only plan additional subgroup analyses in rare cases when there is both a strong theoretical reason to think that there might be differences in effect sizes across subgroups, and for which there is some evidence of differential response from other studies. The panel could also consider seeking community input on the most important group differences to consider but this is likely to generate a long list of potential tests so it would be critical to avoid then conducting many underpowered tests that have a high chance of producing biased results. We encourage panels to follow (judicious) recommendations laid out by various methodologists regarding when and how to determine who is more or less likely to benefit from a given intervention (e.g., Burke et al., 2015; Fisher et al., 2017).

Importantly, panels need to provide full details and transparency about the characteristics of the samples included in the studies that were reviewed so readers can consider how well the results of studies are likely to extend to a given client with their various intersecting identities. Moreover, there is an obvious need for rigorous well-powered studies to inform how well our generally efficacious treatments work with a broad range of marginalized and underserved populations. Finally, it is worth noting that there was some disagreement among our working group members about how strict the limits should be on when to draw

conclusions from studies that were conducted with populations that are often not well represented in clinical trials. In particular, there were different perspectives on the wisdom of including studies that were likely underpowered and had high risk of bias but could provide some empirical data on outcomes for a population that was otherwise not adequately represented in the literature (e.g., in the example provided earlier, the working group varied in whether they felt it would help or hurt to include small cohort studies of prolonged exposure conducted with Black patients). The proposal outlined here attempts to balance the different concerns with an emphasis on not inadvertently denying high-quality care that could be helpful to understudied and underserved groups.

Rationale for Review 2's Descriptive Review of Widely Used Treatments Not Included in Review 1

There are a number of treatments that are widely used in clinical practice and have some empirical support, but the studies of those treatments do not meet the rigorous standards required for inclusion in a guideline's systematic review based on current best practices (psychodynamic treatments, for instance, have not been included in some guideline recommendations for this reason). This may be because blinding did not occur in the ideal way or some aspect of the protocol was not standardized, etc. The challenge is that the field then doesn't learn what we know and don't know about these widely used treatments, so we provide no guidance to clinicians or patients about approaches that are a major part of current practice.

It is risky to make systematic recommendations when the evidence is insufficient or not sufficiently high-quality to support those recommendations, but we can say what we do and do not know by providing a summary of current knowledge with the appropriate caveats about the quality of evidence available. Thus, including a review of a small number of treatments that were not covered in the more traditional systematic review (Review 1) WHEN the treatment is both widely used and the studies are selected based on systematically applied inclusion and exclusion criteria can greatly enhance the clinical utility of guidelines.

This second review would be limited to what is known about efficacy and could include a range of designs, including pre-post, RCTs, and cohort studies. It will be critical to pre-specify the interventions, comparisons, and outcomes that will be extracted, plan evidence tables so the review does not ‘cherry pick’ which results are emphasized, and more broadly, carefully detail the inclusion/exclusion criteria for the types of studies to be reviewed.

Given the expected variability in study designs, doing a meta-analysis will not be an appropriate way to synthesize the data, so this review would not be addressing questions of magnitude and precision of effects with these studies. Rather, because the evidence would by definition not meet the rigorous criteria to be included in the first review, this review would be used to characterize the state of the field (more akin to a scoping review) and guide future research trials. The synthesis of the evidence could follow the PRISMA reporting guidelines (Liberati et al., 2009) and the Synthesis Without Meta-analysis (SWiM) extension (Campbell et al., 2020). Importantly, statements about the pattern or direction of effects would need to be made with considerable caution, and it would need to be clear that these statements do not lead to systematic recommendations for clinical care.

As with Review 1, we encourage the clinical practice guideline panel to determine the list of treatments to review in Review 2 *a priori*. Again, during the scoping process, the panel could solicit community input about which treatments are most widely used to address emotion regulation difficulties and conduct the searches on that list of treatments. For those widely used treatments that do not end up meeting the standards for inclusion in Review 1, they could be evaluated in this second review if they meet the predetermined inclusion and exclusion criteria (see Appendix 2 for sample tools and guidelines that can help evaluate quality and risk for bias in a variety of study designs). It will be critical to differentiate the systematic recommendations following from the first review of efficacy data that is likely to be low in bias from this more descriptive review that should not include recommendations.

Rationale for Review 3's Descriptive Review of Data on Change Processes

We want to know not only what treatments work but also how they work. This is critical for improving care, especially when a treatment is not progressing well or when we are considering using components of a treatment package in complex cases with high comorbidity, etc. Yet, the empirical literature on change processes or principles typically lacks the high-quality trials that *allow for causal inferences* (e.g., the RCTs that are included in systematic reviews of efficacy).

Rather than say nothing about what is known about change processes or just provide a brief description based on expert consensus, we propose conducting a systematic review of high-quality observational data. Studies would be included only if they meet criteria indicating the study accounts for confounds and biases (see below). As with the proposal for Review 2, this review would lead to a summary of the current knowledge with the appropriate caveats about the quality of evidence available but would not lead to recommendations for clinical care.

Review 3 could focus on a subset of change principles from the recent review by Castonguay and colleagues (2019). The selected principles were considered by the working group: 1) to be the most relevant to emotion regulation, AND 2) to have the most research evidence available to evaluate their association with emotion regulation outcomes. In particular, Review 3 could concentrate on the following five change principles:

- Client process principles: 1) resistance/reactance (i.e., client/patient expressions of intolerance or rejection of the therapist's suggestions that may reflect emotion dysregulation)
- Therapy relationship principles: 2) alliance/bond (i.e., purposeful collaboration and affective bond or rupture-repair that can serve as proxies for mutual or co-regulation)
- Therapist intervention principles: 3) self-understanding (e.g., reappraisal and meaning making to support emotion regulation), 4) emotional experiencing (such that higher levels of experiencing indicate greater emotion differentiation, meaning making, and thus

regulation of emotional experience), and 5) behavior change (i.e., behavioral strategies designed to shift or manage emotion: see Table 1 for more info on these principles' relevance for emotion regulation).

The review would include studies that examined any of these five change principles in the context of treatments that appear to have some success changing emotion regulation (based on Reviews 1 and 2), but not any study that examined the five change principles outside the context of treating emotion regulation. For instance, evidence tied to the self-understanding change principle underlying the effects of emotion-focused therapy for depression would be included but not a study examining self-understanding in the context of motivational interviewing for substance use disorders.

Note the working group fully recognizes that there are many possible change processes or principles that could be of interest, and this is intentionally a brief list (to manage scope of the review and make it more feasible that a three-review guideline could be achieved). As with Reviews 1 and 2, we encourage the clinical practice guideline development panel to seek community input and expert consultation before finalizing the list of change principles that would be reviewed but do encourage the panel to keep the list small to avoid guidelines taking many years to produce. Also, it will be critical to further specify the specific questions of interest for the review and include a published review protocol prior to conducting the searches (but after establishing a search strategy). This will help ensure the review is systematic and reduce bias.

Along these lines, questions for Review 3 should be formulated according to a model such as Sample, Phenomenon of Interest, Design, Evaluation, Research type (SPIDER, Cooke et al., 2012). Considerable work will be required to determine terms related to the principles of interest. Questions should be limited to research where change processes have been related to overall outcome. Research could be evaluated using appraisal checklists for a variety of observational studies from Cochrane or CASP (see Appendix B for sample tools and guidelines

that can help evaluate quality and risk for bias in a variety of study designs). And, as recommended in review 2, SWiM could be applied as a synthesis method.

NOTE. It will be important to look for opportunities to be inclusive throughout the process so diverse voices are involved in making decisions about which subpopulations, if any, are examined separately in Review 1, which treatments are included in Review 2, which change principles are examined in Review 3, etc. This will include having good representation in the composition of the guideline panel and when seeking expert input.

Managing Scope By Clustering Target Populations

Given we are proposing three review components, we propose to manage scope by focusing on a subset of problem areas at a time. Although we ultimately want to review the literature for many different problem areas given the transdiagnostic focus, this could be done in stages so each guideline would focus on a theoretically coherent subset of literature bases. In this way, we propose a compromise that avoids trying to review every possible problem area in one guideline (a process that would take many years and result in an indigestible, exceedingly long guideline) but also avoids looking only within one disorder at a time, thereby moving far away from the desired transdiagnostic focus.

There are multiple ways to determine clusters of problem areas that could be grouped together for a given guideline and the panel will need to decide how it wants to predefine subsets of problem areas. The working group thought a few different approaches were reasonable so we include the various options here. One option would be to consider empirical evidence for coherence or comorbidity (e.g., could review comorbidity data, or examine classification systems like the Hierarchical Taxonomy of Psychopathology; HiTOP, Kotov et al., 2017). Alternatively, clustering could be guided by theoretical/conceptual links regarding underlying mechanisms across problem areas or could simply be based on pragmatic considerations tied to what problem areas have good quality studies to review.

The working group proposed the following problem areas --at either diagnostic or subthreshold levels-- to be included in clinical practice guidelines on emotion regulation (though, as noted, not all in one guideline). As with other aspects of this proposal, we offer this list as a starting point and encourage the clinical practice guideline panel to seek community and expert input on the most important problem areas to include, with the proviso that there needs to be sufficient *high-quality literature that examines emotion regulation outcomes for this problem area* before it makes sense to conduct a systematic review on that problem area.

- Mood disorders or high depressive symptoms
- Anxiety disorders or high anxious symptoms
- Suicidal behavior/self-injurious behaviors
- Borderline personality disorder (BPD) or high PBD symptoms
- Substance use problems
- Eating disordered behavior
- Complicated grief
- Aggression/conduct problems
- Trauma
- Dissociative behavior
- Rumination
- Procrastination

Analogously, although we eventually want to know what the literature indicates across the lifespan, a first guideline could focus on younger and middle-aged adults, followed by a later one on children and adolescents, and one on older adults. It is hoped that these reviews could be facilitated by the Agency for Healthcare Research and Quality.

Preliminary Evaluation of the Feasibility of Reviewing Efficacy Studies that Include Emotion Regulation Outcome Measures

One challenge facing the working group was how to determine whether there was likely to be sufficient data available to review to produce a guideline. This is a difficult question to answer with confidence when the scoping and search parameters have not been finalized, and we encourage a future clinical practice guideline panel to consider this issue carefully when determining the specific search terms and study inclusion/exclusion criteria that would be outlined in a review protocol. It was nonetheless important to make an initial determination in order to address the critical question of feasibility of doing a guideline. To this end, the working group decided to use two prior reviews of therapies intended to improve emotion regulation (both conducted in the past five years) as a source of available studies (Gratz et al., 2015; Moltrecht et al., 2020). Although this approach is certainly imperfect given a future clinical practice guideline panel is likely to choose a different search process and use different inclusion/exclusion criteria for study selection, it is an efficient way to get a reasonable estimate of whether there is a body of literature to review.

Given Review 1 is expected to apply the most stringent criteria for study inclusion (i.e., for a study to be deemed a sufficiently high-quality test of efficacy that would allow for causal inferences without high risk of bias) so that recommendations can be made, we systematically coded each of the studies included in the two reviews to determine: 1) Was the study design an RCT? 2) Was the targeted problem area one of the problems from our recommended list (see Managing scope by clustering target populations)? 3) Did the article examine change in emotion regulation using one of the measures we identified in Table 2?

A study had to meet all three of these criteria for us to assume it would likely provide data for a future Review 1. Based on these two review papers, 28 studies qualified (14 studies addressed adult populations, 14 studies addressed children/adolescents; see Appendix A). The working group felt this was sufficient evidence to suggest that there would likely be enough data

to review to warrant developing a clinical practice guideline. (Note we did not do a similar exercise for Reviews 2 and 3 because those reviews are designed to be descriptive summaries of what is known in the area and would not lead to recommendations.)

Needs for Future Research

Over the course of its work, our working group identified several important areas for future research based on gaps in the literature we identified:

- Well-powered tests are needed that use study designs that allow for causal inferences, have low risk of bias, and that:
 - Examine change in emotion regulation outcomes across a range of therapeutic approaches:
 - There are a number of treatments that are widely used and conceptually expected to target and change emotion regulation, but which have not been well tested in high-quality clinical trials that reduce risk of bias and directly measure emotion regulation outcomes.
 - Even among well-tested treatments, the efficacy studies often lack measures of emotion regulation outcomes, and the measured outcomes rarely capture the multi-faceted nature of emotion regulation or use methods other than self-report.
 - Are designed to examine subgroup differences (treatment heterogeneity) for studies with established evidence of effectiveness:
 - Many studies are not sufficiently powered to properly examine subgroup differences (e.g., differences related to demographic variables, such as ethnicity, race, gender identity, etc.).
 - Examine principles of change that test specific mechanisms that account for change in emotion regulation outcomes:

- We know little about whether interventions are effective and less about why and how interventions might achieve their effects, especially when trying to draw causal inferences (rather than examining associations).
 - Isolate the necessary and sufficient therapy elements to change emotion regulation and the therapy techniques that are especially effective to implement those elements. We need to move beyond treatment packages so treatment plans can be more precise and personalized to each client's particular needs.
- Research to better define and conceptualize emotion regulation along the full spectrum of healthy and clinical populations. We encourage development of more shared terminology and taxonomies across basic and clinical researchers from different schools/theoretical traditions so that it can be easier to translate basic research findings to clinical practice and vice versa, and so similarities and differences across diverse treatment approaches can be better defined.
- Enhanced clarity and standardization on how to identify individuals with emotion regulation problems with attention to diagnostic and psychometric considerations:
 - There is currently no clear agreement among researchers and clinicians on how best to identify individuals with emotion regulation difficulties and what thresholds to use to indicate severity. An approach that can work across problem areas/disorders is needed

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Footnotes

¹ We recognize there are many concepts and terms related in some way to problems in emotion regulation in the literature on psychopathology. For example, there are important literatures tied to: *defense mechanisms (security operations)*, intrapersonal and interpersonal ways by which one manages emotions and conflict (see Cramer, 2008); *experiential avoidance*, a strategy commonly used to reduce anxiety by avoiding anxiety-arousing experiences (see Roemer et al., 2008); *dissociation*, a mechanism by which we separate from emotional material by selectively in-attending to anxiety-linked stimuli and splitting off traumatic material (see Steele et al., 2016); *self-splits*, patient behaviors, states or markers that indicate a conflict between two aspects of the self and an interruption of one aspect of the self by another (see Greenberg et al., 1993); *parasuicidal and self-destructive behavior*, self-harming actions that regulate and soothe painful emotions (see Linehan, 1993), among others.

² We also recognize there are many concepts and terms related in some way to emotion regulation in the literature on psychotherapy intervention. For example, *decentering or distancing*, efforts to develop patients' ability to become more aware of automatic processes through some form of guided discovery (see Beck et al., 1979); *mindfulness (acceptance- or compassion-based) training*, to develop self-observation with some detachment to allow one to experience emotions without over-reaction to or judgment of intense negative emotions or disturbing thoughts (see Hayes et al., 1999, Roemer et al., 2008, and Segal et al., 2002); *affect identifying and labeling (granularity)*, techniques that foster awareness of the physiological concomitants of feelings, abilities to appropriately label them and to linguistically share them with others (see Niles et al., 2015); *exposure and exploration*, strategies to desensitize the individual to overwhelming affective states and reduce the intensity of intensely negative emotional states via repeated exposure (see Barlow, 2002; Craske et al., 2014); *cognitive reappraisal (and restructuring)*, techniques that strengthen the ability to use executive functions to down-regulate emotional reactivity (see Gross, 1998); *mentalization (mentalized affectivity,*

second-order representation), techniques that encourage self-reflective capacities and the ability to put thoughts and feelings into linguistic form so that one feels located and understood (see Fonagy et al, 2004); *experiencing (differentiation, complexity)*, techniques that encourage differentiation and tolerance to experience a wide range of emotions without becoming emotionally overloaded (see Gendlin, 1996; Greenberg, 2016); *attentional distraction*, techniques that are used to strengthen emotional regulation by shifting the focus from something disturbing and emotionally intense to something less anxiety arousing (see Gross, 1998); *expressive suppression (response modulation)*, any behavioral technique or exercise that inhibits the experience of an emotion (see Gross, 1998); *stimulus control or modification*, basic techniques of behavioral therapy used to shape and manage emotional tolerance (see Linehan, 1993), among others.

Appendix A

Studies from Gratz et al. (2015) and Moltrecht et al. (2020) That Meet Criteria for Likely Inclusion in a Future Review 1

Table A1

Adult and Older Adult Population Studies from Gratz et al. (2015) That Meet Criteria for Likely Inclusion in a Future Review 1 (N = 14)

Author, year	Population	Treatment*	Emotion Regulation measure(s) (Author, year)
Berking et al., 2013	Age: M = 46.44 years Condition: Major Depressive Disorder Country: Germany	CBT-Emotion Regulation Skills Training (ERT) vs. CBT	Emotion Regulation Skills Questionnaire (Berking et al., 2008)
Cloitre et al., 2002	Age: M = 34 years Condition: PTSD due to childhood abuse Country: USA	Skills Training in Affect and Interpersonal Regulation (STAIR) + Modified PE vs. WL [†]	Toronto Alexithymia Scale (Bagby et al., 1994) Generalized Expectancy for Negative Mood Regulation Scale (Cantanzaro & Mearns, 1990)
Goldin et al., 2012	Age: 21-55 years Condition: Social Anxiety Disorder Country: USA	Individual CBT vs. WL	Emotion Regulation Questionnaire (Gross & John, 2003)
Gratz & Gunderson, 2006	Age: M = 33 Condition: BPD w/ self-harm Country: USA	Acceptance-Based Emotion Regulation Group Intervention vs. TAU	Acceptance and Action Questionnaire (Hayes et al., 2004) Difficulties in Emotion Regulation Scale (Gratz & Roemer, 2004)
Gratz et al., 2014	Age: M = 33.3 Condition: BPD Country: USA	Emotion Regulation Group Therapy vs. TAU	Acceptance and Action Questionnaire - II (Bond et al., 2011) Difficulties in Emotion Regulation Scale (Gratz & Roemer, 2004)

Author, year	Population	Treatment*	Emotion Regulation measure(s) (Author, year)
Juarascio et al., 2013	Age: M = 26.74 years Condition: Eating disorders Country: USA	Acceptance and Commitment Therapy + TAU vs. TAU	Acceptance and Action Questionnaire - II (Bond et al., 2011) Difficulties in Emotion Regulation Scale (Gratz & Roemer, 2004) Drexel Defusion Scale (Forman et al., 2012) Philadelphia Mindfulness Scale (Cardaciotto et al., 2008)
Masson et al., 2013	Age: M = 42.8 Condition: Binge Eating Disorder Country: Canada	Dialectical Behavior Therapy for Binge Eating Disorder vs. WL	Difficulties in Emotion Regulation Scale (Gratz & Roemer, 2004)
Morton et al., 2012	Age: M(ACT+TAU) = 35.6 years; M(TAU) = 34 years Condition: BPD Country: Australia	Acceptance and Commitment Therapy + TAU vs. TAU	Acceptance and Action Questionnaire - II (Bond et al., 2011) Affective Control Scale (Williams et al., 1997) Difficulties in Emotion Regulation Scale (Gratz & Roemer, 2004) Five-Facet Mindfulness Questionnaire (Baer et al., 2006)
Neacsiu, Eberle et al., 2014	Age: M = 18 years or older Condition: Emotion dysregulation (depression/anxiety) Country: USA	Dialectical Behavior Therapy-Skills training vs. Activities-based support group	Difficulties in Emotion Regulation Scale (Gratz & Roemer, 2004)
Neacsiu, Lungu et al., 2014	Age: 18-45 years Condition: BPD Country: USA	Dialectical Behavior Therapy vs. Community Treatment By Experts	Acceptance and Action Questionnaire - II (Bond et al., 2011)

Author, year	Population	Treatment*	Emotion Regulation measure(s) (Author, year)
Robins et al., 2012	Age: M = 46.25 years Condition: Psychological distress Country: Canada	Mindfulness-Based Stress Reduction vs. WL	Affective Control Scale (Williams et al., 1997) Difficulties in Emotion Regulation Scale (Gratz & Roemer, 2014) Five-Facet Mindfulness Questionnaire (Baer et al., 2006) Ruminative Response Scale (Nolen-Hoeksema & Morrow, 1991) Spielberger Anger Expression Scale (Spielberger et al., 1985)
Roemer et al., 2008	Age: M = 33.59 years Condition: GAD Country: USA	Acceptance Based Behavior Therapy vs. WL	Acceptance and Action Questionnaire (Hayes et al., 2004) Mindful Attention Awareness Scale (Brown & Ryan, 2003)
Safer et al., 2010	Age: 18 and older Condition: Binge Eating Disorder Country: USA	Dialectical Behavior Therapy-Binge Eating Disorder vs. Active comparison group therapy	Difficulties in Emotion Regulation Scale (Gratz & Roemer, 2004) Generalized Expectancies for Negative Mood Regulation Scale (Catanzaro & Mearns, 1990)
Wonderlich et al., 2014	Age: M = 27.3 Condition: Bulimia Nervosa Country: USA	Integrative cognitive-affective therapy vs. CBT for Bulimia Nervosa	Difficulties in Emotion Regulation Scale (Gratz & Roemer, 2004)

Note. *All are randomized controlled trials; [†]TAU = Treatment as Usual; WL = Waitlist

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Table A2

Child/Adolescent Population Studies from Moltrecht et al. (2020) that meet criteria for likely inclusion in a future Review 1 (N = 14)

Author, year	Population	Treatment*	Emotion Regulation measure(s) (Author, year)
Afshari et al., 2014	Age: 9-13 years Condition: Separation Anxiety Disorder Country: Iran	Group Emotion-focused CBT vs. Group CBT vs. Control	Children's Sadness Management Scale (Zeman et al., 2001) Children's Worry Management Scale (Zeman et al., 2010)
Atkinson & Wade, 2016	Age: Adolescent Condition: Eating disorder Country: Australia	Mindfulness vs. WL	Five-Facet Mindfulness Questionnaire (Baer et al., 2006)
Azrin et al., 2001	Age: 12-17 years Condition: Comorbid CD + SUD or Sub. dependence OR Comorbid Sub. dependence and ODD Country: USA	Individual Cognitive Problem Solving vs. active control (Family Behavioral Therapy)	Social Problem Solving Inventory-Revised (D'Zurilla et al., 2002)
Dingle & Fay, 2017	Age: 18-25 years Condition: Depression, anxiety, stress Country: Australia	Tuned-In vs. WL	Difficulties in Emotion Regulation Scale (Gratz & Roemer, 2004)
Fitzpatrick et al., 2005	Age: Mean 19.02 years Condition: Suicidal ideation (depression) Country: USA	Problem-solving vs. health education	Social Problem Solving Inventory-Revised (D'Zurilla et al., 2002)
Hides et al., 2011	Age: 16-25 years Condition: Comorbid depression and substance misuse Country: Australia	CBT/Motivational Interviewing vs. Standard Alcohol and Other Drug Care	Coping Inventory for Stressful Situations (Cosway et al., 2000)

Author, year	Population	Treatment*	Emotion Regulation measure(s) (Author, year)
Jacobs et al., 2016	Age: Adolescents Condition: Major Depressive Disorder Country: USA and UK	Rumination-focused CBT vs. Assessment Only	Ruminative Response Scale (Nolen-Hoeksema & Morrow, 1991)
Latimer et al., 2003	Age: Mean 16.07 years Condition: SUD Country: USA	Integrated Family and Cognitive-Behavioral Therapy vs. Psychoeducation	Social Problem Solving Inventory-Revised (D'Zurilla et al., 2002)
Livheim et al., 2015	Age: 12-18 years (AUS); 14-15 years (SWE) Condition: Depression and stress Country: Australia and Sweden	Acceptance and Commitment Therapy vs. Control (individual support)	Mindful Attention Awareness Scale (Brown & Ryan, 2003)
Meiser-Stedman et al., 2017	Age: 8-17 years Condition: PTSD Country: UK	Cognitive Therapy for PTSD vs. WL	Trauma-Related Rumination Items (Meiser-Stedman et al., 2014)
Schuppert et al., 2012	Age: 14-19 years Condition: BPD Country: The Netherlands	Emotion Regulation Training vs. TAU	Life Problems Inventory (Rathus et al., 2015)
Slee et al., 2008	Age: 15-35 years Condition: Deliberate Self-Harm/Borderline Personality Disorder (BPD) Country: The Netherlands	CBT vs. TAU	Difficulties in Emotion Regulation Scale (Gratz & Roemer, 2004)
Smith et al., 2015	Age: 12-16 years Condition: Depression Country: UK	Computerized CBT vs. WL	Child Response Style Questionnaire (Meiser-Stedman et al., 2007)
Suveg et al., 2018	Age: 7-12 years Condition: Anxiety Country: USA	Emotion-Focused CBT vs. CBT	Emotion Regulation Checklist (Shields & Cicchetti, 1997)

Note. *All are randomized controlled trials; †TAU = Treatment as Usual; WL = Waitlist

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Appendix B

The following links may be useful as the guideline development group considers the criteria to guide inclusion/exclusion of studies for reviews 2 and 3:

- a. Review re confounding in observational studies:

<https://www.annualreviews.org/doi/pdf/10.1146/annurev-clinpsy-032816-045030>

- b. Cochrane ROBINS-I checklist:

<https://sites.google.com/site/riskofbiastool//welcome/home>

- c. CASP cohort study checklist:

http://docs.wixstatic.com/ugd/dded87_e37a4ab637fe46a0869f9f977dacf134.pdf

- d. CASP case control checklist:

http://docs.wixstatic.com/ugd/dded87_63fb65dd4e0548e2bfd0a982295f839e.pdf

- e. CASP qualitative checklist:

http://docs.wixstatic.com/ugd/dded87_29c5b002d99342f788c6ac670e49f274.pdf